Seunguk Kang

Department of Chemistry, Korea Advanced Institute of Science and Technology (KAIST)
Daejeon, 34141, Republic of Korea

drkd319@kaist.ac.kr | +82-10-4168-2616 | Google Scholar | GitHub

Education

Korea Advanced Institute of Science and Technology (KAIST) 2022 – Present
B.S. in Chemistry; expected graduation: February 2028 • Daejeon, Republic of Korea

Korea Science Academy (KSA) 2019 – 2021
High School Diploma – Chemistry • Busan, Republic of Korea

Publications and Manuscripts

[J3] Hyoju Choi, **Seunguk Kang**, Mu-Hyun Baik*. "Mechanistic Origins of Selectivity in Brønsted Acid-Assisted Solid-State [2+2] Photocycloaddition." *Manuscript in preparation.*

[J2] Patrick Gross, Hyoju Choi, Wesley Pullara, Hoyoung Im, Khang Ung, **Seunguk Kang**, Mu-Hyun Baik*, Simon B. Blakey*. "Enantioselective Olefin 1,2-Arylamination Catalyzed by a Planar Chiral Indenyl-Rhodium(III) Complex." *ACS Catal.* **2026**, 16(4), 3453–3463.

[J1] Taegyun Hong, Changjae Lee, Yeongseo Bak, Geonhyeong Park, Hongju Lee, **Seunguk Kang**, Tae-Hyun Bae, Dong Ki Yoon*, Jesse G. Park*. "On-Demand Tunable Electrical Conductance Anisotropy in a MOF-Polymer Composite." *Small* **2024**, 20(18), 2309469. *Featured on the front cover.*

Research Experience

Intelligent Chemistry Lab, KAIST
Dec. 2025 – Present
Undergraduate Researcher • Advisor: Prof. Woo Youn Kim

Neural Free-Energy Estimation for Molecular Ensembles
- Investigated a neural free-energy estimator for organic molecules using learned transport between prior and molecular ensemble distributions and nonequilibrium work statistics.
- Adapted and retrained the molecular conformer generative model and implemented forward/reverse sampling kernels, path-probability-based work calculations, and Bennett acceptance ratio estimation.

Full-Atom Antibody Design with Antigen Flexibility
- Investigated full-atom antibody design that accounts for antigen flexibility and induced fit with a Riemannian Diffusion Bridge-based E(3)-equivariant generative framework.
- Developed auxiliary training losses and validation tools for steric and stereochemical validity.

AD-Agent: Mechanism-Aware Drug Repurposing for Alzheimer's Disease
- Investigated mechanism-guided drug repurposing for Alzheimer's disease using multi-agent reasoning.
- Implemented Open Targets retrieval functions to integrate mechanistic and clinical with traceable sources.

Baik Group, KAIST
Jan. 2024 – Dec. 2025
Undergraduate Researcher • Advisor: Prof. Mu-Hyun Baik

Brønsted Acid-Assisted Solid-State [2+2] Photocycloaddition [J3]
Collaboration: Prof. Tehshik P. Yoon (University of Wisconsin–Madison)
- Investigated how the local molecular environment and water influence stereoselectivity and reactivity in solid-state photocycloaddition via DFT and QM/MM calculations.

Enantioselective Olefin 1,2-Arylamination with Chiral Indenyl–Rh Catalysts [J2]
Collaboration: Prof. Simon B. Blakey (Emory University)
- Investigated indenyl effects and how N-protecting groups modulate nitrogen nucleophilicity and chemoselectivity between arylamination and aziridination via DFT calculation.

Soft Material Assembly Group, KAIST
Jun. 2023 – Dec. 2023

Undergraduate Researcher • Advisor: Prof. Dong Ki Yoon

Tunable Electrical Conductance Anisotropy in MOF–Polymer Composites [J1]

- Investigated tunable electrical conductance anisotropy through electric-field alignment of conductive MOFs within a polymer matrix.

Choi Group, KSA
Apr. 2019 – Dec. 2021

Student Researcher • Advisor: Prof. Eun Young Choi

Activation and Adsorption Studies of MOFs and Zeolites

- Developed mild activation protocols for fragile porphyrin-based MOFs to improve gas adsorption and conducted high-vacuum gas and ion exchange experiments on zeolite single crystals.

Biocompatible Metal–Tartrate MOFs and Their Antibacterial Properties

Singapore International Collaborative Research Program • Team Leader

- Synthesized Zn-, Co-, and Ag-tartrate MOFs through rapid room-temperature self-assembly and investigated their antibacterial properties.

Research Methods & Technical Skills

Computational Methods & Scientific Software

- Generative models (diffusion, flow, stochastic interpolants); agent tools and API integration.
- DFT (ORCA, ADF), MD (Amber), QM/MM; statistical mechanics, nonequilibrium work relations.
- Python (PyTorch, RDKit, NumPy, pandas); Linux/Bash, Slurm, Git.

Experimental Chemistry & Characterization

- MOF synthesis and activation; inert-atmosphere and high-vacuum techniques; PXRD, SEM/EDS, TGA, UV-Vis, IR; surface area and porosimetry analysis.

Selected Presentations

Chiral Indenyl Rh-Catalyzed Enantioselective Olefin 1,2-Arylamination

- Korean Chemical Society Spring Conference (poster, **2025**); Chemistry Department Workshop (oral, **2024**).

Biocompatible MOFs Developed by Facile Ultra-Fast Room Temperature Synthesis

- Korean Chemical Society Future Chemists Research Conference (poster, **2021**).

Improvement of Gas Adsorption Efficiency of Fragile Porphyrin-based MOFs

- Thailand International Science Fair (oral, **2021**); Korean Polymer Society Autumn Conference (poster, **2020**).

Optimizing Synthesis Conditions of PPF-4: Effect of Acid Addition on Crystal Growth

- Japan Super Science Fair (oral, **2020**).

Honors and Awards

Best Participant Award and **2nd Prize, IR Pitching Contest**, Global Entrepreneurship Summer School 2024

Talent Award of Korea, Ministry of Education 2021
one of 100 national recipients recognized for future leadership

Academic Excellence Award in Chemistry, Korea Science Academy 2020, 2021

Gold Prize, World Invention Creativity Olympic (WICO) 2017

Minister's Award, Korean Youth Invention Idea Contest, Ministry of Science and ICT 2017